

SUPPLEMENTARY INFORMATION

Conformational Chirality of Single-Crystal Covalent

Organic Frameworks

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Materials

All solvents and reagents were used as received without further purification unless otherwise specified. *p*-Toluenesulfonic acid monohydrate (purity $\geq 98\%$) and Tetrahydrofuran (AR, purity $\geq 99\%$) were purchased from Aladdin Co. Tetrakis(4-aminophenyl)methane (TAM) (purity $\geq 98\%$), 1, 4-phthalaldehyde (purity $\geq 98\%$), were purchased from Jilin Chinese Academy of Sciences-Yanshen Technology Co., Ltd. Dodecanoyl-*L*-alanine (C_{12} -*L*-AlaA, purity $\geq 99\%$) was purchased from Adamas-beta®. Tetradecanoyl-*L*-alanine (C_{14} -*L*-AlaA, purity $\geq 97\%$) was purchased from Konosience. Hydrochloric acid (36% ~38%), acetone (AR, purity $\geq 99\%$) and sodium hydroxide (AR, purity $\geq 96\%$) were purchased from Xilong Scientific and Beijing Tongguang fine chemical Company, respectively.

Circular dichroic spectra (CD): The CD spectrometer is produced by JASCO Company, model J-815, equipped with 150W air-cooled xenon lamp, and the measured wavelength range is 200 nm - 800 nm with a scanning speed of 100 nm min⁻¹. The UV-visible absorption spectrum is tested synchronously. COF-300_{*L*-cp} and COF-300_{*D*-cp}: The 5 mg sample was dispersed in 30 mL water to obtain colloids for testing; COF-300_{*L*-V}: The sample was applied to quartz flakes after drying for testing; The variable temperature CD spectrum is tested by adding a programmed temperature control attachment where the test range is 25 °C - 95 °C, and the heating rate is 10 °C min⁻¹. The temperature fluctuation is less than 0.1 °C. The samples were maintained at target temperature for 5 minutes for each test.

Continuous precession electron diffraction tomography (cPEDT): The COF-300_{*L*-op}, COF-300_{*L*-cp} and COF-300_{*D*-cp} were carefully dispersed onto a copper grid (R2/1.Quantifoil). The grid was then plunge-frozen in the liquid ethane with the assist of VitrobotTM. A Fischione2550 cryo-transfer tomography holder was used to keep the grid in low temperature when the grid was transferred into the TEM. cPEDT data were acquired using a JEOL 2100-Plus transmission electron microscope equipped with a MerlinEM-4S direct electron detector operating at an acceleration voltage of 200 kV

and maintained at 100 K throughout the entire data collection. All the data were reconstructed and visualized with program *REDp*²⁶ and processed using *XDS*²⁷ with the aid of *Instamatic*²⁸ for batch processing. The structure solution and refinement were conducted with the *SHELXT*²⁹ and *SHELXL*³⁰ program respectively.

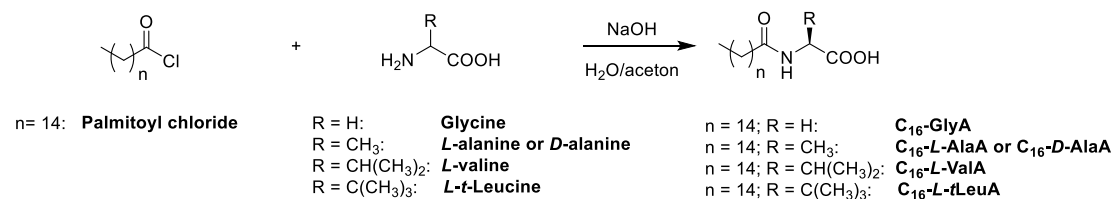
Powder X-ray diffraction (PXRD): All hydrated COF-300 samples were first soaked in water, air-dried naturally, and then sealed in a capillary tube. Their powder X-ray diffraction data were collected on PANalytical Empyrean diffractometer equipped with Cu K α radiation source at 45 kV and 40 mA in transmission mode. The PXRD data of the rest of samples were collected in reflection mode. Pawley and Rietveld refinements of these experimental PXRD profiles were conducted by *TOPAS-V5*³¹ software. All the structural analysis in this paper is established based on the accurate structural model obtained via Rietveld refinement.

***In-situ* PXRD during CO₂ adsorption:** The *in-situ* PXRD patterns during gas adsorption process were collected on a Rigaku SmartLab diffractometer equipped with cryogenic holder and MicrotracBEL Max adsorption instruments. The sample were placed in the chamber with Be windows and then activated at 120 °C vacuum for 12 h before measurement. The PXRD patterns were collected simultaneously with the adsorption isotherm.

Gas and vapor adsorption: Adsorption isotherms of nitrogen and water vapor were measured by MicrotracBEL Max II adsorption instruments. The sample was activated at 120 °C vacuum for 12 hours prior to testing.

Section 1. Synthesis procedures and analysis

Section 1.1. Synthesis procedures of amino-acid derivatives



General procedure A: Amino-acid derivatives were synthesized according to a reported procedure¹⁷. Briefly, we dissolved amino acid (0.1 mol) and sodium hydroxide (0.1 mol, 4.0 g) in a mixed solvent of acetone (80 mL) and water (120 mL), then added aliphatic acid chloride (0.1 mol) and sodium hydroxide aqueous solution (0.2 mol NaOH in 30 mL of water) slowly and alternatively at 0 °C. After stirring for two hours, the pH of the solution was adjusted to 1 with 6 mol L⁻¹ hydrochloric acid, and then the crude product was obtained by filtering. The product was washed three times with 250 mL of petroleum ether to obtain amino acid derivatives as white solids.

Section 1.2 Synthesis of chiral COF-300 and COF-304

General procedure B: To synthesize the chiral COFs, we dissolved 0.4 mmol of amino-acid derivative templates in 25 mL of water along with 0.2 mol L⁻¹ sodium hydroxide solution (2 mL). The mixture was stirred at 50 °C for 10 minutes to form an emulsion. Then, 0.2 mol L⁻¹ hydrochloric acid (2 mL) was added, followed by the addition of terephthalaldehyde derivatives 0.2 mmol. Subsequently, a solution of tetrakis(4-aminophenyl)methane (TAM, 38 mg, 0.1 mmol) in 0.2 mol L⁻¹ PTSA (2 mL) was introduced. After one hour reaction, the product was collected by filtration and washed three times with tetrahydrofuran and water to yield chiral COFs.

Synthesis of chiral COF-300_L or COF-300_D: To synthesize the chiral COFs, we dissolved 0.4 mmol of amino-acid derivative templates (palmitoyl-L-alanine (C₁₆-L-AlaA) for the synthesis of COF-300_L, palmitoyl-D-alanine (C₁₆-D-AlaA) for the synthesis of COF-300_D) in 25 mL of water along with 0.2 mol L⁻¹ sodium hydroxide solution (2 mL). The mixture was stirred at 50 °C for 10 minutes to form an emulsion. Then, 0.2 mol L⁻¹ hydrochloric acid (2 mL) was added, followed by the addition of

benzaldehyde (BDA, 26.8 mg, 0.2 mmol). Subsequently, a solution of tetrakis(4-aminophenyl)methane (TAM, 38 mg, 0.1 mmol) in 0.2 mol L⁻¹ PTSA (2 mL) was introduced. After one hour reaction, the product was collected by filtration and washed three times with tetrahydrofuran and water to yield chiral COFs.

Synthesis of chiral COF-304: To synthesize the chiral COFs, we dissolved 0.4 mmol of amino-acid derivative templates (palmitoylglycine (C₁₆-GlyA) for the synthesis of COF-304, palmitoyl-*L*-alanine (C₁₆-*L*-AlaA) for the synthesis of COF-304_L, and palmitoyl-*D*-alanine (C₁₆-*D*-AlaA) for the synthesis of COF-304_D) in 25 mL of water along with 0.2 mol L⁻¹ sodium hydroxide solution (2 mL). The mixture was stirred at 50 °C for 10 minutes to form an emulsion. Then, 0.2 mol L⁻¹ hydrochloric acid (2 mL) was added, followed by the addition of 2,3-dihydroxyterephthalaldehyde (2,3-DHTA), 33.2 mg, 0.2 mmol. Subsequently, a solution of tetrakis(4-aminophenyl)methane (TAM, 38 mg, 0.1 mmol) in 0.2 mol L⁻¹ PTSA (2 mL) was introduced. After one hour reaction, the product was collected by filtration and washed three times with tetrahydrofuran and water to yield chiral COFs. The detailed experimental conditions summarized in Table S6.

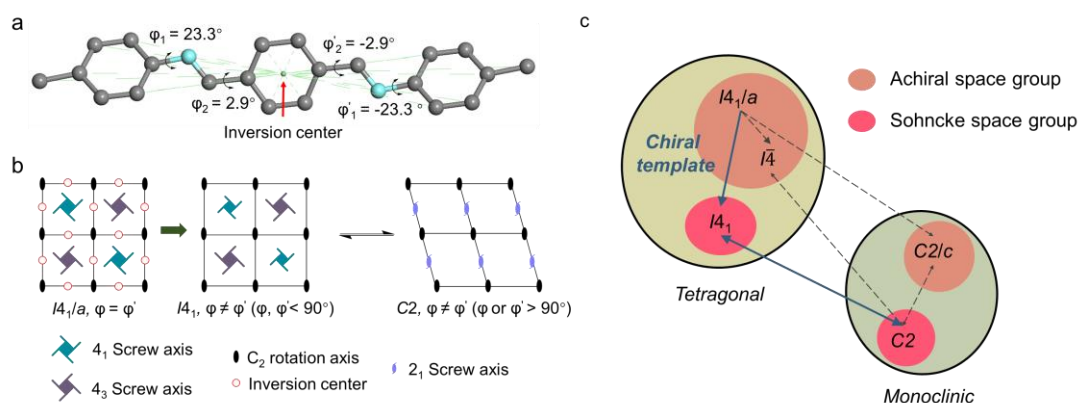


Figure S1. Chiral conformation and crystal symmetry of COF-300. **a**, Linker of achiral COF-300 with symmetric conformation. **b**, Symmetric operations of the $I4_1/a$, $I4_1$ and $C2$ space groups. **c**, Conformational changes may result in alterations to the space group. The solid arrows depict the findings reported in this work.

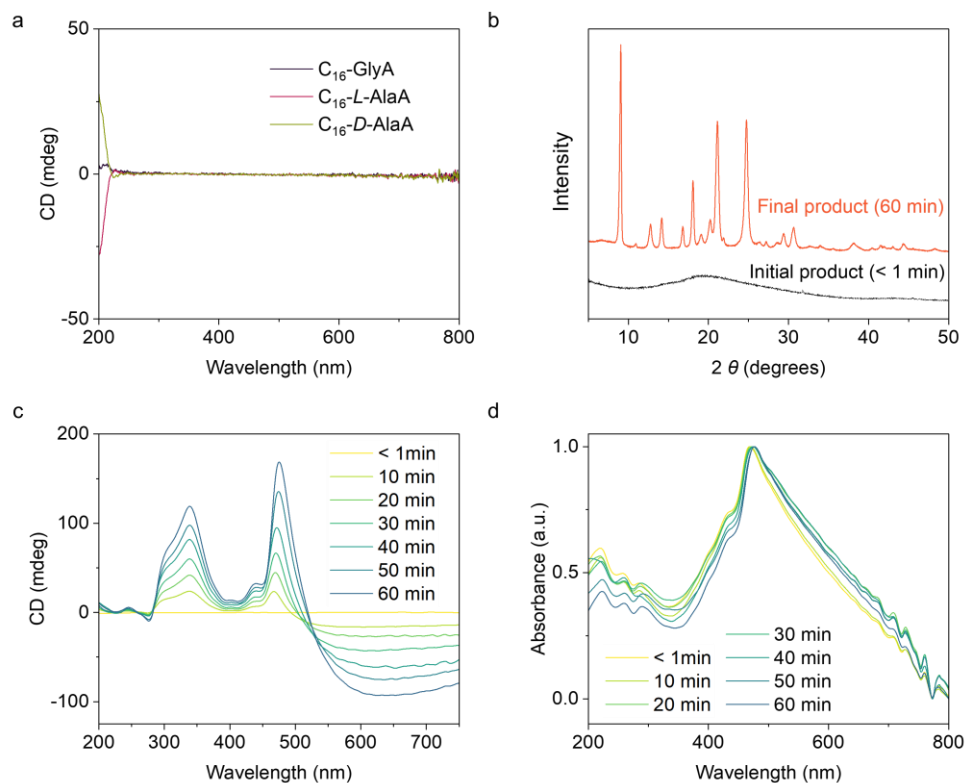


Figure S2. Observation of the formation of conformationally chiral COF-300_L. **a**, Circular dichroic (CD) absorption spectra of C₁₆-GlyA, C₁₆-L-AlaA and C₁₆-D-AlaA micelles. **b**, Powder X-ray diffraction (PXRD) patterns of initial product (< 1 min) and final product (60 min). **c**, **d**, CD (**c**) and UV-visible (UV-vis) (**d**) absorption spectra of the products after the removal of amino-acid derivatives at various reaction times.

Time-course study of the growth process of chiral COF-300_L: In accordance with the *General procedure B*, seven reactions were concurrently conducted using 0.4 mmol of amino-acid derivative (C₁₆-L-AlaA) and 0.2 mmol of terephthalaldehyde derivatives (BDA). Subsequently, the resulting products were filtered sequentially at intervals of < 1 min, 10 min, 20 min, 30 min, 40 min, 50 min, and 60 min. Each filtered sample was washed successively with tetrahydrofuran and water to remove the templates, followed by natural air drying in preparation for subsequent tests.

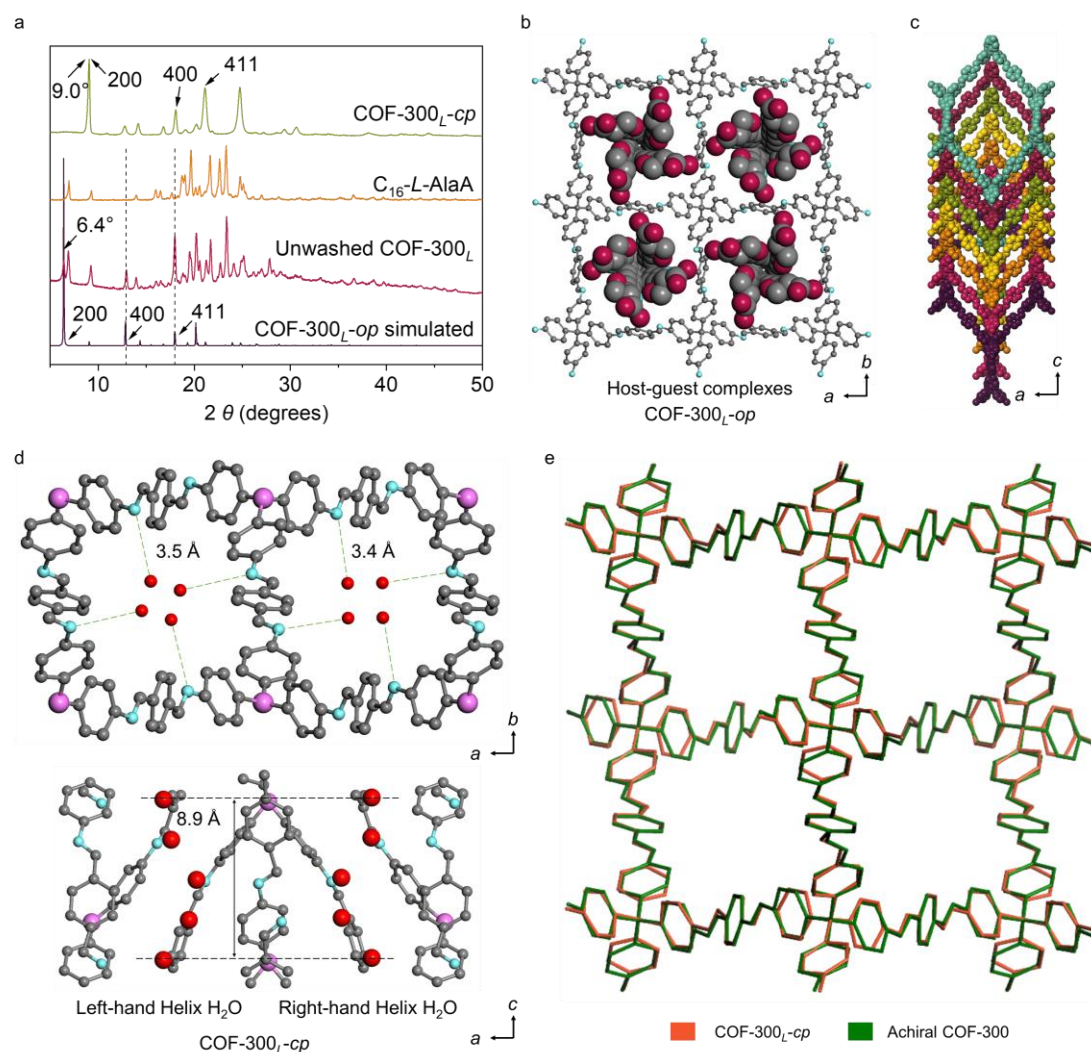


Figure S3. Structural of conformational chiral COFs before and after the removal of chiral template. **a**, PXRD patterns of the unwashed COF-300_L, C₁₆-L-AlaA and COF-300_L-cp together with the simulated PXRD pattern of COF-300_L-op. **b**, Crystal structure of COF-300_L-op whose pores are filled with disordered chiral amino-acid derivatives. **c**, Seven-fold interpenetrated structure of the COF-300_L-op. **d**, Crystal structure of COF-300_L-cp. **e**, Crystal structure overlay of COF-300_L-cp and achiral COF-300.

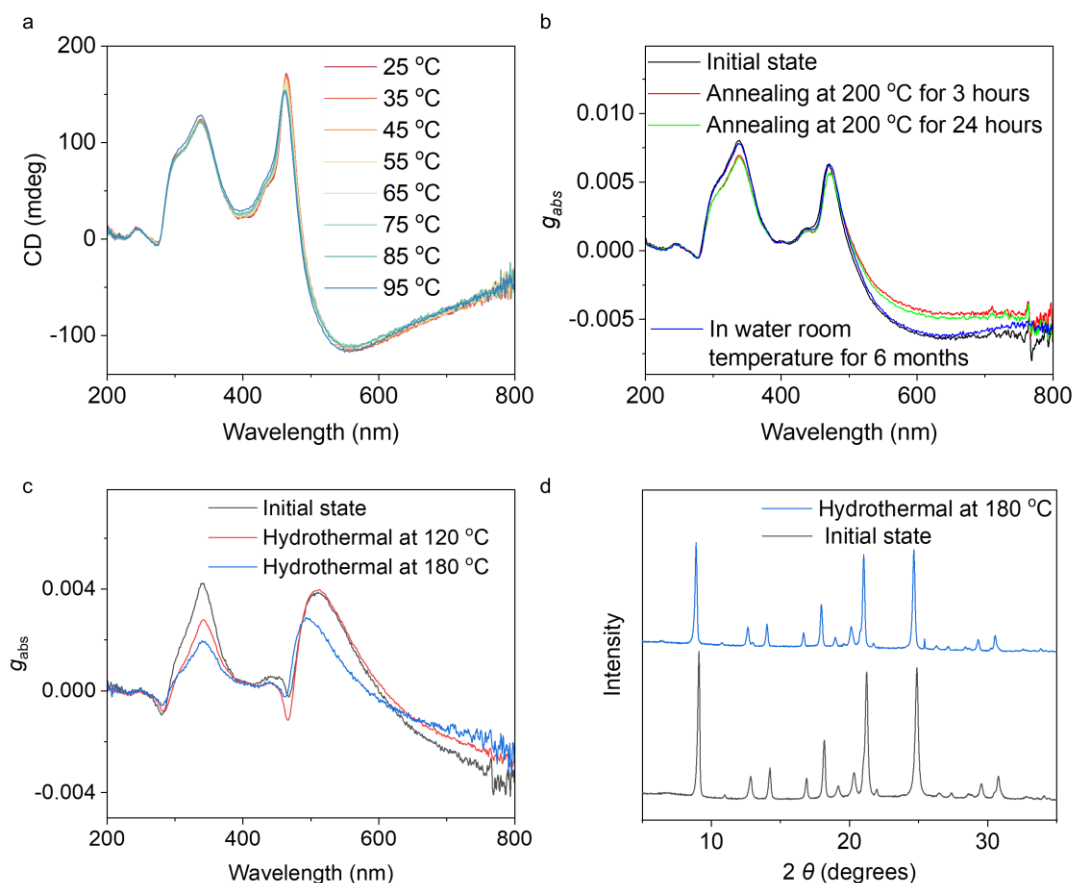


Figure S4. Stability of conformational chiral COF-300. **a**, Variable temperature CD absorption spectra from 25 °C to 95 °C. **b**, **c**, CD absorption spectra of chiral COF-300 after various treatments. **d**, PXRD patterns of chiral COF-300 initial state and hydrothermal at 180 °C.

Stability of conformational chiral COFs: Thermal annealing: Place 10 mg COF-300_L into a 25 ml crucible and heat it in a drying oven at 200 °C for either 3 hours or 24 hours. After cooling, disperse the solid in water for circular dichroism spectroscopy testing. Hydrothermal treatment: Disperse 10 mg COF-300_L in 30 mL of water and place it in a hydrothermal reactor. Heat at either 120 °C or 200 °C in an oven for 6 hours. After cooling, filter the solid and use the filtered material directly for PXRD testing. Re-disperse the solid in water for circular dichroism spectroscopy testing.

Table S1 | Dihedral angles corresponding to chiral conformations in linker.

	φ_1	φ'_1	φ_2	φ'_2
300_L-op	-28.9 (R)	34.6 (S)	-3.9 (R)	15.0 (S)
300_L-cp	-8.5 (R)	-19.0 (R)	13.6 (S)	-3.0 (R)
300_D-cp	-22.6 (R)	43.4 (S)	-4.8 (R)	-10.7 (R)
300_L-V	-3.3 (R)	162.6 (R)	-7.4 (R)	175.5 (R)

Table S2 | Characteristic dimensions of chiral COF-300 structures.

		Helix		Helix H ₂ O	
		Left-hand	Right-hand	Left-hand	Right-hand
300_L-op	Helix diameter	12.9 Å	13.2 Å	--	--
	Pitch	53.1 Å		--	
300_L-cp	Helix diameter	9.8 Å	9.9 Å	2.4 Å	2.1 Å
	Pitch	62.5 Å		8.9 Å	
300_D-cp	Helix diameter	9.9 Å	9.8 Å	2.6 Å	1.8 Å
	Pitch	62.5 Å		8.9 Å	
300_L-V	Helix diameter	--	--	--	--
	Pitch	56.6 Å		--	

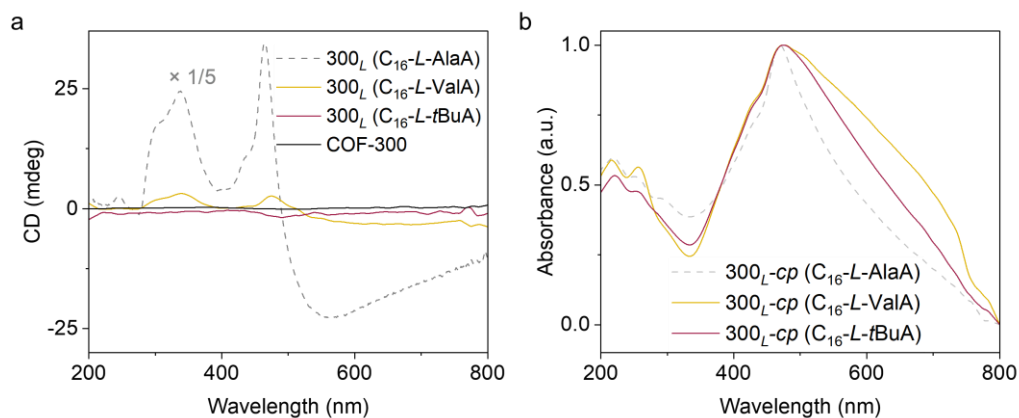


Figure S5. Steric hindrance effect of chiral amino-acid derivative. **a, b**, CD (**a**) and UV-visible (**b**) absorption spectra of COF-300_L-cp synthesized using amino-acid derivatives with various steric hindrance of functional groups (C₁₆-L-AlaA, C₁₆-L-ValA, C₁₆-L-*t*BuA). These COF-300_L samples were synthesized according to the *General procedure B*, with detailed experimental conditions summarized in Table S4.

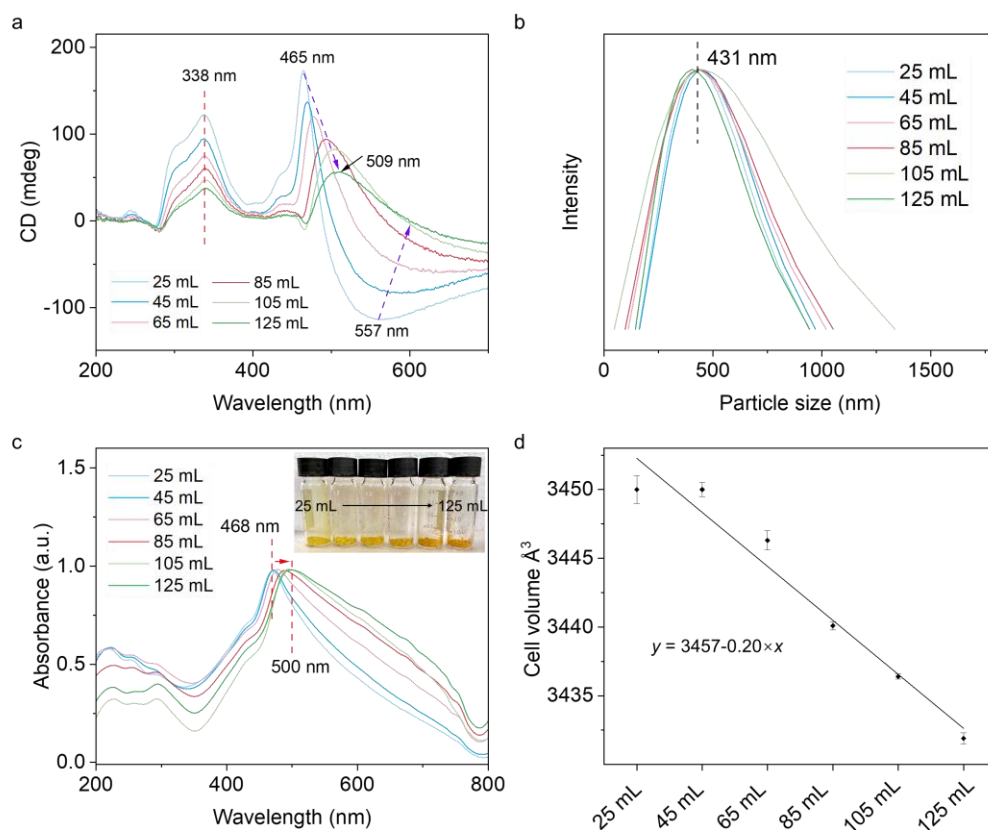
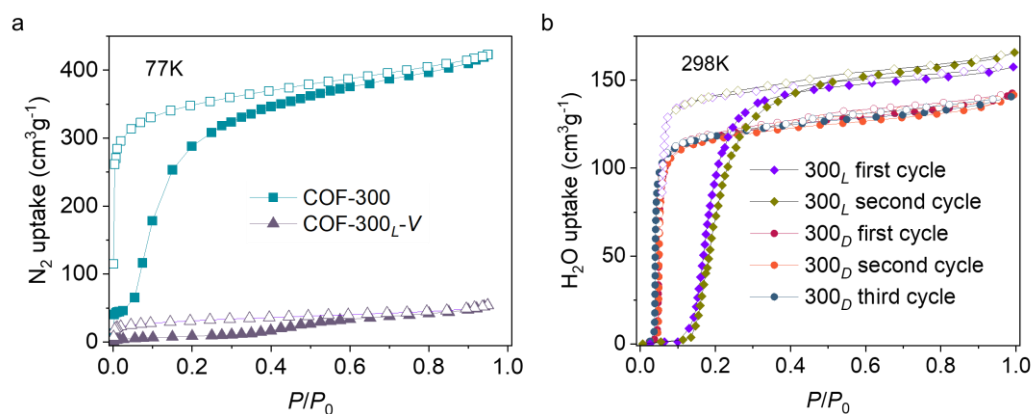


Figure S6. Effect of concentration of chiral amino-acid derivatives. **a**, **b**, **c**, **d**, CD spectra (**a**), particle size (**b**), UV-vis absorption spectra (**c**) and unit cell volume (**d**) of COF-300_L-cp were analyzed under various water concentrations (25 mL, 45 mL, 65 mL, 85 mL, 105 mL, and 125 mL).

Study on the effect of concentration: Following the *General procedure B*, a 0.4 mmol quantity of C₁₆-L-AlaA was utilized, while keeping other conditions consistent. The solvent water was added in increments of 25 mL, 45 mL, 65 mL, 85 mL, 105 mL, and 125 mL respectively. After one hour reaction, the resulting mixture was washed separately with tetrahydrofuran and water. Subsequently, the sample was dried in air naturally for subsequent testing. The detailed experimental conditions summarized in Table S5.

Table S3. Absorption dissymmetry factor of chiral materials.

Name	g_{abs}	Reference
Chiral COF(^R COF)	2×10^{-2}	Angew. Chem. Int. Ed. 61, e202113979 (2022).
Chiral COF(CityU-7)	-2.3×10^{-4}	Adv. Mater. 36, 2306414 (2024).
Chiral COF(CityU-8)	4.5×10^{-4}	
Tp(Δ)-COF	$5.27 \times 10^{-4} \sim 6.39 \times 10^{-4}$	J. Am. Chem. Soc. 146, 13201-13209 (2024).
Tp(Λ)-COF	$-5.06 \times 10^{-4} \sim -9.52 \times 10^{-4}$	
4-<i>P</i>	1.24×10^{-3}	J. Am. Chem. Soc. 143, 4661-4667 (2021).
6-<i>P</i>	1.06×10^{-2}	
(<i>R</i> -NEA) ₃ Ge ₂ I ₇ ·H ₂ O	3.75×10^{-4}	Angew. Chem. Int. Ed. 62, e202309600 (2023).
(<i>S</i> -NEA) ₃ Ge ₂ I ₇ ·H ₂ O	-3.96×10^{-4}	
Chiral COF-300	$4.0 \times 10^{-3} \sim 1.2 \times 10^{-2}$	This work

**Figure S7.** Adsorption and desorption cycles of gas/vapor in COFs. **a**, Adsorption and desorption isotherms of N₂ at 77 K for COF-300 and COF-300_L-V. **b**, Adsorption and desorption isotherms of water vapor at 298 K for COF-300_L-V and COF-300_D.

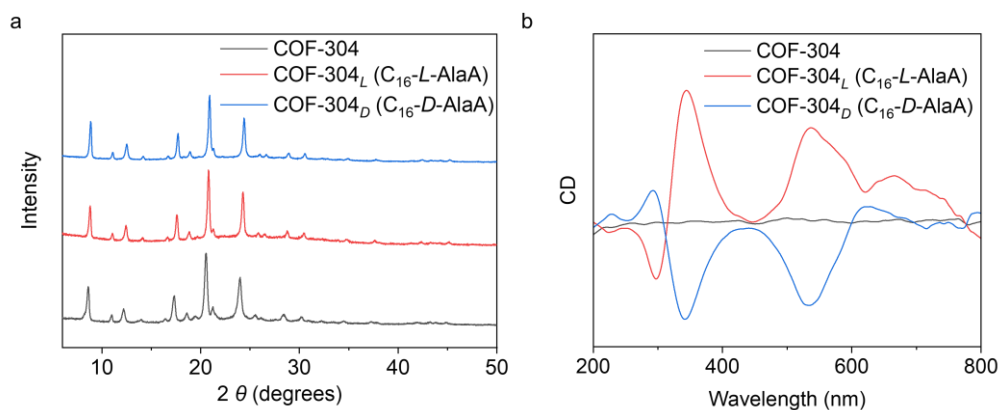


Figure S8. Conformational chirality of COF-304. PXRD patterns (a) and CD absorption spectra (b) of achiral COF-304, and COF-304_L, and COF-304_D. We utilized the chiral templates C₁₆-L-AlaA and C₁₆-D-AlaA to synthesize the conformational chiral COF-304_L and COF-304_D through the condensation of TAM and 2,3-dihydroxyterephthalaldehyde (2,3-DHTA) (Methods). PXRD analysis confirmed high crystallinity for all the samples. CD spectroscopy revealed nearly mirror-image signals for COF-304_L and COF-304_D. In contrast, COF-304 synthesized without chiral templates did not display any CD signals.

Table S4. Summary of synthetic chiral COF-300 at amino-acid derivative micelles.

Amino acid derivatives	TAM 0.1 mmol	BDA 0.2 mmol	NaOH 0.2 mol L ⁻¹	HCl 0.2 mol L ⁻¹	PTSA 0.2 mol L ⁻¹	Water
C ₁₂ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₄ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₆ -L-ValA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₆ -L- <i>t</i> BuA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₆ -D-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL

Table S5. Summary of synthetic chiral COF-300 at different concentrations.

Amino acid derivatives	TAM 0.1 mmol	BDA 0.2 mmol	NaOH 0.2 mol L ⁻¹	HCl 0.2 mol L ⁻¹	PTSA 0.2 mol L ⁻¹	Water
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	25 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	45 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	65 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	85 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	105 mL
C ₁₆ -L-AlaA	38 mg	26.8 mg	2 mL	2 mL	2 mL	125 mL

Table S6. Summary of synthetic chiral COF-304 at amino-acid derivative micelles.

Amino acid derivatives	TAM 0.1 mmol	2,3-DHTA 0.2 mmol	NaOH 0.2 mol L ⁻¹	HCl 0.2 mol L ⁻¹	PTSA 0.2 mol L ⁻¹	Water
C ₁₆ -GlyA	38 mg	33.3 mg	2 mL	0 mL	5 mL	25 mL
C ₁₆ -L-AlaA	38 mg	33.3 mg	2 mL	0 mL	5 mL	25 mL
C ₁₆ -D-AlaA	38 mg	33.3 mg	2 mL	0 mL	5 mL	25 mL

Section 2. 3D electron diffraction (3D ED) analysis

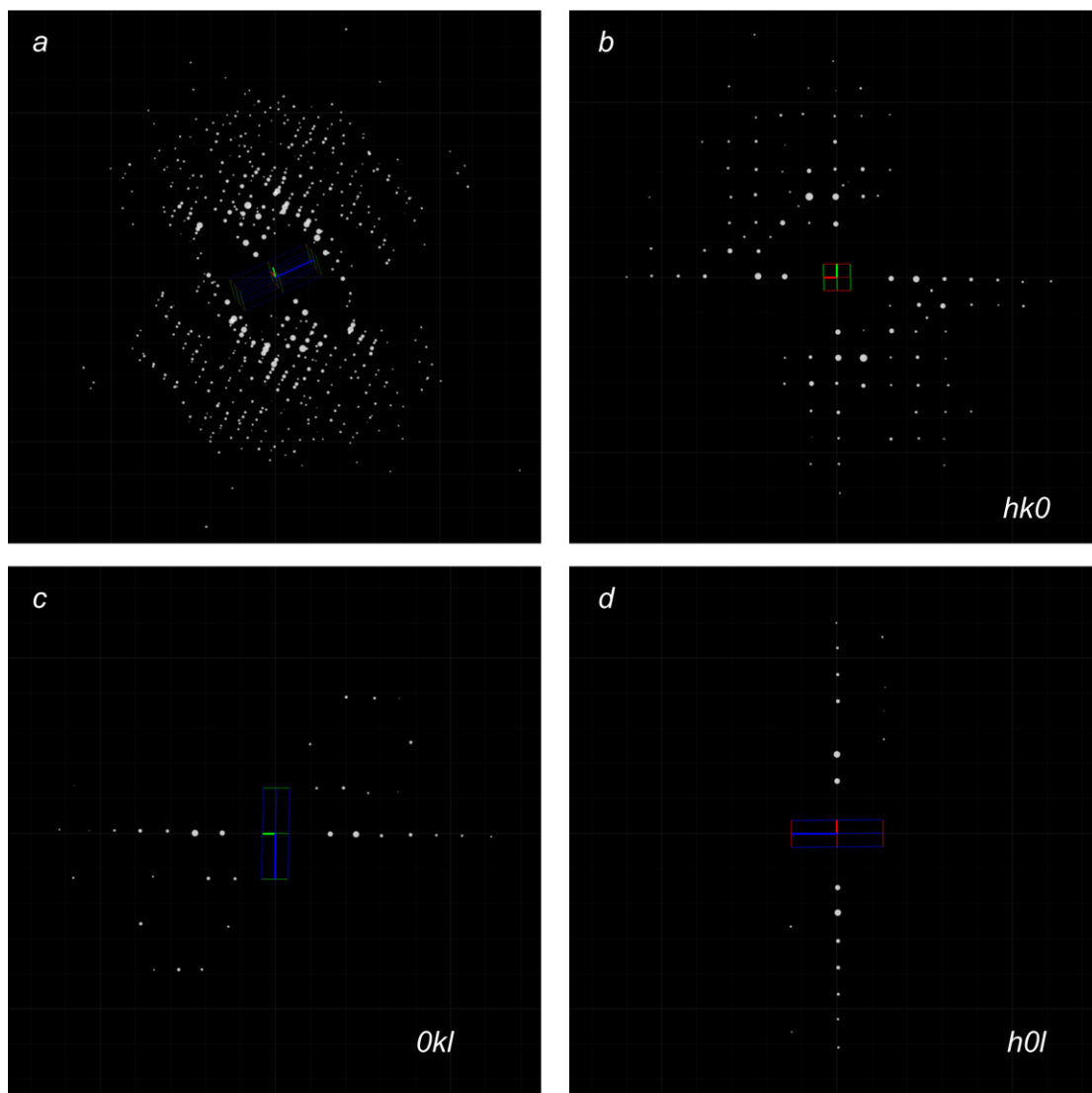


Figure S9. Reconstructed 3D reciprocal lattice of COF-300_{L-op}. a, Overview of reconstructed lattice of COF-300_{L-op}. b, c, d, Selected slices cut from the reconstructed 3D reciprocal lattice of COF-300_{L-op} showing the *hk0* (b), *h0l* (d), *0kl* (c). The reflection conditions can be obtained as: *hkl*: $h + k + l = 2n$, *hk0*: $h + k = 2n$; *0kl*: $k + l = 2n$; *h00*: $h = 2n$.

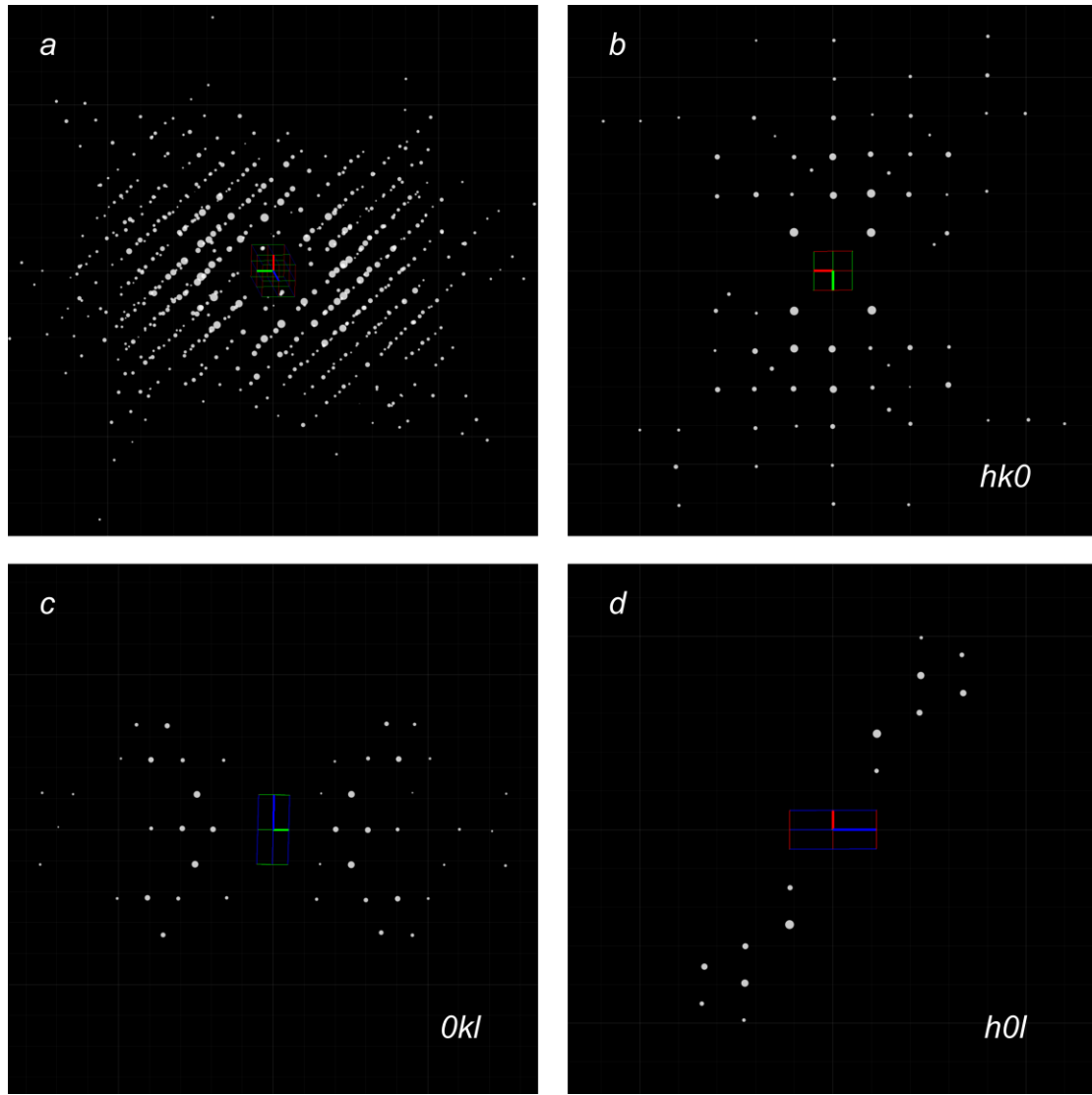


Figure S10. Reconstructed 3D reciprocal lattice of COF-300_L-cp. a, Overview of reconstructed lattice of COF-300_L-cp. b, c, d, Selected slices cut from the reconstructed 3D reciprocal lattice of COF-300_L-cp showing the $hk0$ (b), $h0l$ (d), $0kl$ (c). The reflection conditions can be obtained as: hkl : $h + k + l = 2n$, $hk0$: $h + k = 2n$; $0kl$: $k + l = 2n$; $h0l$, $h + l = 2n$.

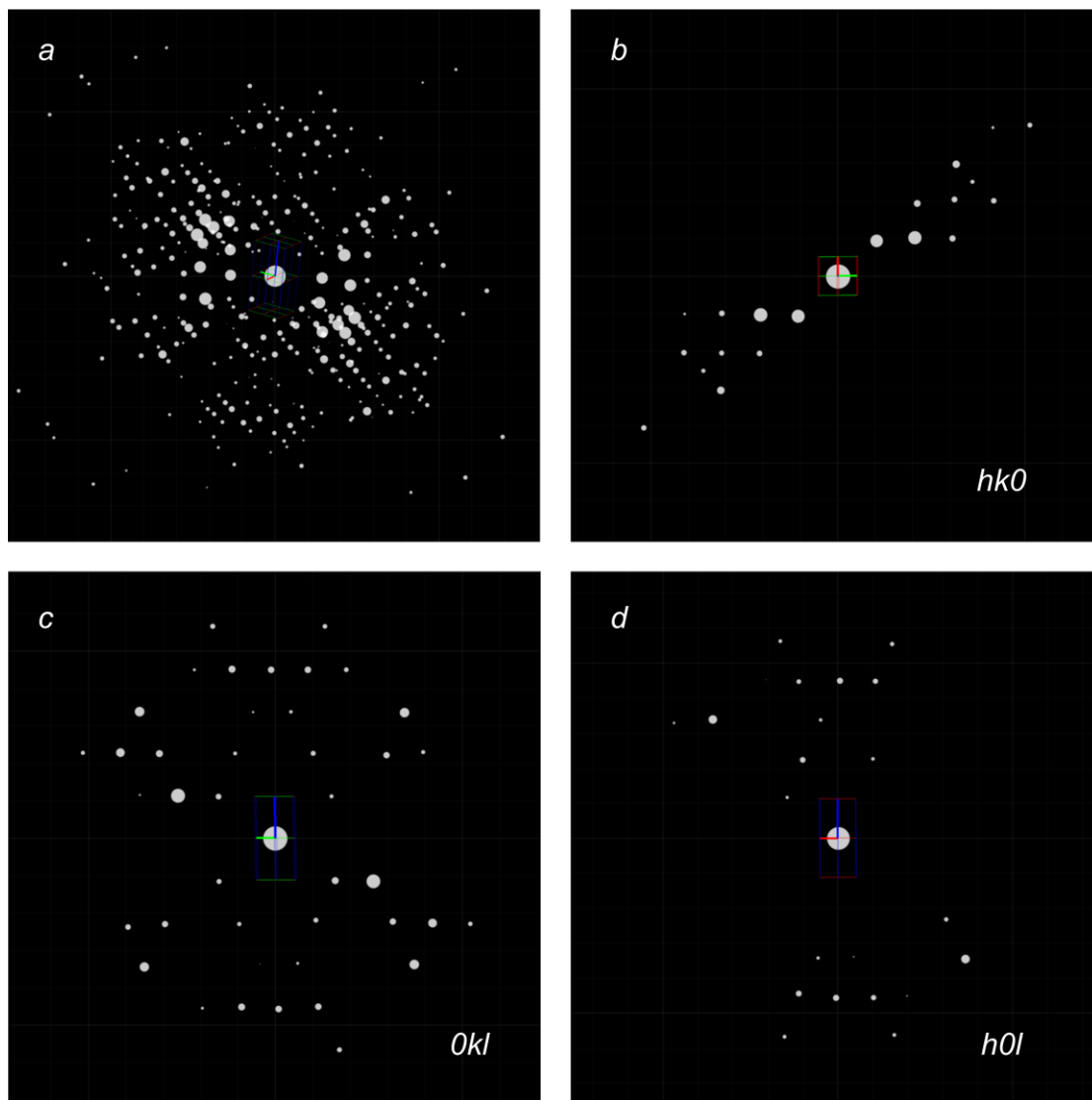


Figure S11. Reconstructed 3D reciprocal lattice of COF-300_D-cp. a, Overview of reconstructed lattice of COF-300_D-cp. b, c, d, Selected slices cut from the reconstructed 3D reciprocal lattice of COF-300_D-cp showing the $hk0$ (b), $h0l$ (d), $0kl$ (c). The reflection conditions can be obtained as: hkl : $h + k + l = 2n$, $hk0$: $h + k = 2n$; $0kl$: $k + l = 2n$; $h0l$: $h + l = 2n$.

Table S7. Crystallographic data and structure refinement details performed on the merged 3D ED dataset of COF-300_{L-op} and COF-300_{L-cp}.

Name	COF-300 _{L-op}	COF-300 _{L-cp}
Molecular formula	C ₄₁ N ₄ H ₂₈	C ₄₁ N ₄ H ₂₈ •4H ₂ O
Data collection method	cPEDT	cPEDT
Temperature / K	100	100
Wavelength / Å	0.0251	0.0251
Crystal system	Tetragonal	Tetragonal
Space group	<i>I</i> 4 ₁ (No. 80)	<i>I</i> 4 ₁ (No. 80)
Unit cell dimensions	$a = b = 26.130(4) \text{ Å}$, $c = 7.5900(15) \text{ Å}$, $\alpha = \beta = \gamma = 90^\circ$	$a = b = 19.946(3) \text{ Å}$, $c = 8.8307(18) \text{ Å}$, $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	5182.3(15)	3513.3(10)
<i>Z</i>	4	4
Resolution / Å	0.8	1
<i>R</i> _{int}	0.267	0.239
Completeness / %	80	89.6
<i>R</i> ₁	0.206	0.179
<i>wR</i> ₂	0.612	0.515

Table S8. Crystallographic data and structure refinement details performed on the merged 3D ED dataset of COF-300_D-cp.

Name	COF-300 _D -cp
Molecular formula	C ₄₁ N ₄ H ₂₈ •4H ₂ O
Data collection method	cPEDT
Temperature / K	100
Wavelength / Å	0.0251
Crystal system	Tetragonal
Space group	<i>I</i> 4 ₁ (No. 80)
Unit cell dimensions	$a = b = 19.430(3)$ Å, $c = 8.8400(18)$ Å, $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3337.3(9)
<i>Z</i>	4
Resolution / Å	1
<i>R</i> _{int}	0.277
Completeness / %	76.5
<i>R</i> ₁	0.164
<i>wR</i> ₂	0.462

Section 3. Powder XRD analysis

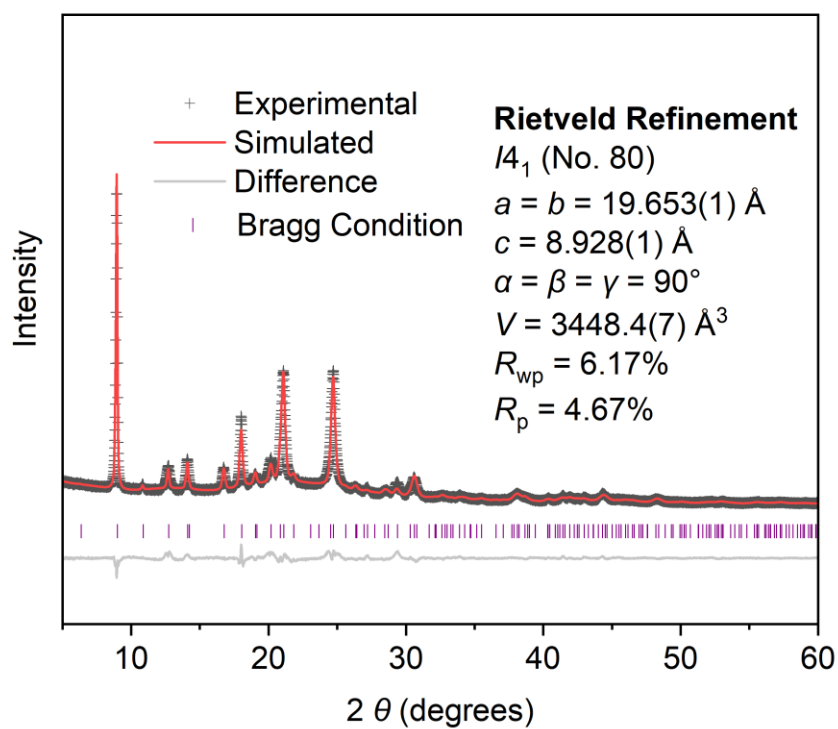


Figure S12. Rietveld refined powder X-ray diffraction pattern of COF-300_L-cp.

Table S9. Summary for the Rietveld refinement results of COF-300_L-cp.

Name	COF-300 _L -cp
Molecular formula	C ₄₁ N ₄ H ₂₈ •4H ₂ O
Temperature / K	Room temperature
Wavelength / Å	1.54056
2 θ range for extraction / °	5 ~ 60
Data collection	Transmission mode
Crystal system	Tetragonal
Space group	<i>I</i> 4 ₁ (No. 80)
Unit cell dimensions	$a = b = 19.653(1)$ Å, $c = 8.928(1)$ Å, $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3448.4(7)
Density / g/cm ³	1.180(3)
<i>R</i> <i>p</i> / %	4.67
<i>R</i> <i>w</i> <i>p</i> / %	6.17
<i>GOF</i>	2.94

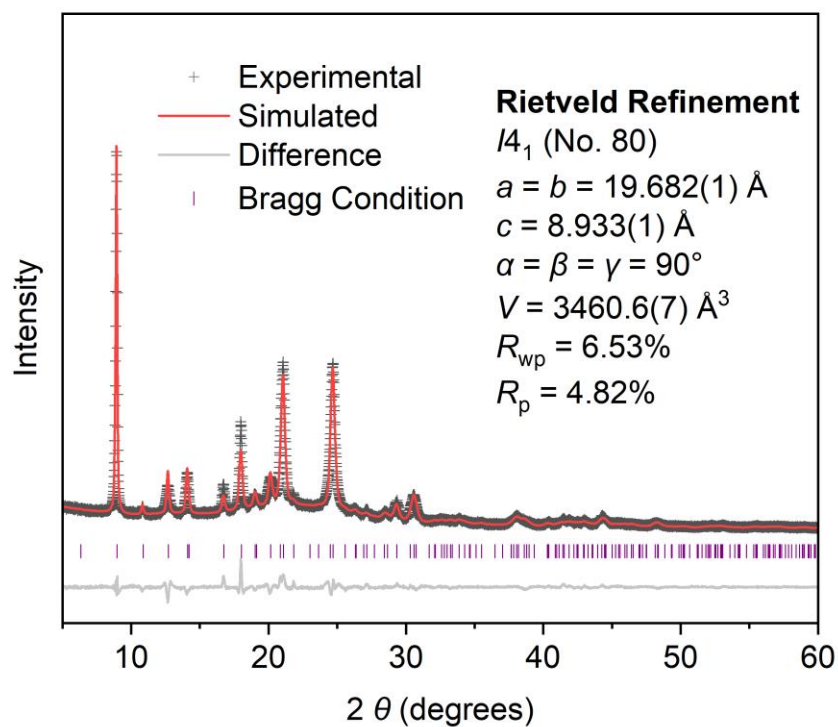


Figure S13. Rietveld refined powder X-ray diffraction pattern of COF-300_D-cp.

Table S10. Summary for the Rietveld refinement results of COF-300_D-cp.

Name	COF-300 _D -cp
Molecular formula	C ₄₁ N ₄ H ₂₈ •4H ₂ O
Temperature / K	Room temperature
Wavelength / Å	1.54056
2 θ range for extraction / °	5 ~ 60
Data collection	Transmission mode
Crystal system	Tetragonal
Space group	<i>I</i> 4 ₁ (No. 80)
Unit cell dimensions	$a = b = 19.682(1) \text{ Å}$, $c = 8.933(1) \text{ Å}$, $\alpha = \beta = \gamma = 90^\circ$
Cell volume / Å ³	3460.6(7)
Density / g/cm ³	1.176(2)
<i>R</i> <i>p</i> / %	4.82
<i>R</i> <i>w</i> <i>p</i> / %	6.53
<i>GOF</i>	2.23

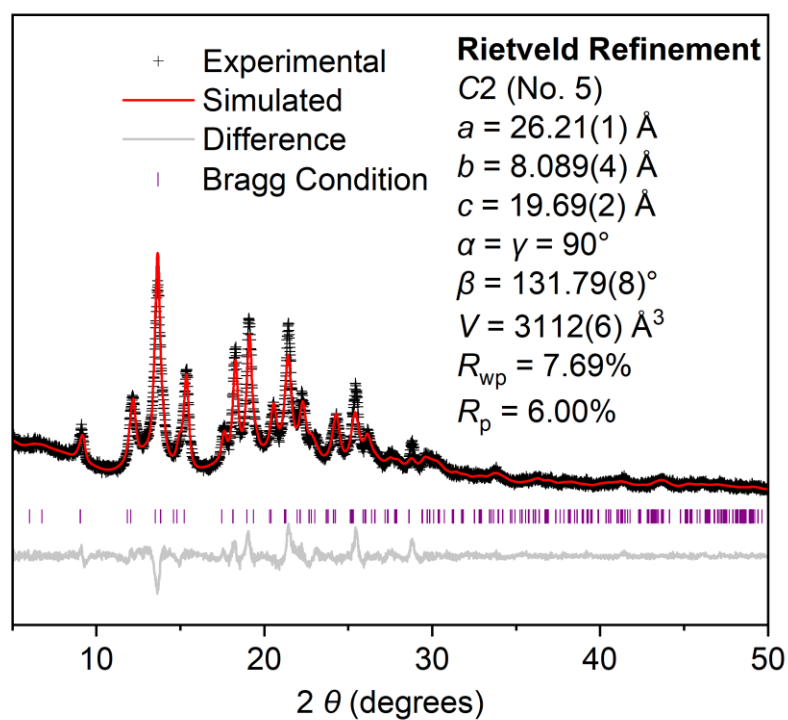


Figure S14. Rietveld refined powder X-ray diffraction pattern of COF-300_L-V.

Table S11. Summary for the Rietveld refinement results of COF-300_L-V.

Name	COF-300 _L -V
Molecular formula	C ₄₁ N ₄ H ₂₈
Temperature / K	Room temperature
Wavelength / Å	1.54056
2 θ range for extraction / °	5 ~ 50
Data collection	Reflection mode
Crystal system	Monoclinic
Space group	C2 (No. 5) *
Unit cell dimensions	$a = 26.21(1) \text{ Å}$, $b = 8.089(4) \text{ Å}$, $c = 19.69(2) \text{ Å}$, $\alpha = \gamma = 90^\circ$, $\beta = 131.79(8)^\circ$
Cell volume / Å ³	3112(6)
Density / g/cm ³	1.171(2)
R_p / %	6.00
R_{wp} / %	7.69
GOF	1.50

* Powder X-ray indexing suggested compatibility with the higher-symmetry space group *Fdd2*. However, due to the chiral signal observed in the sample, the maximal subgroup *C2* of *Fdd2* was identified as the actual space group.

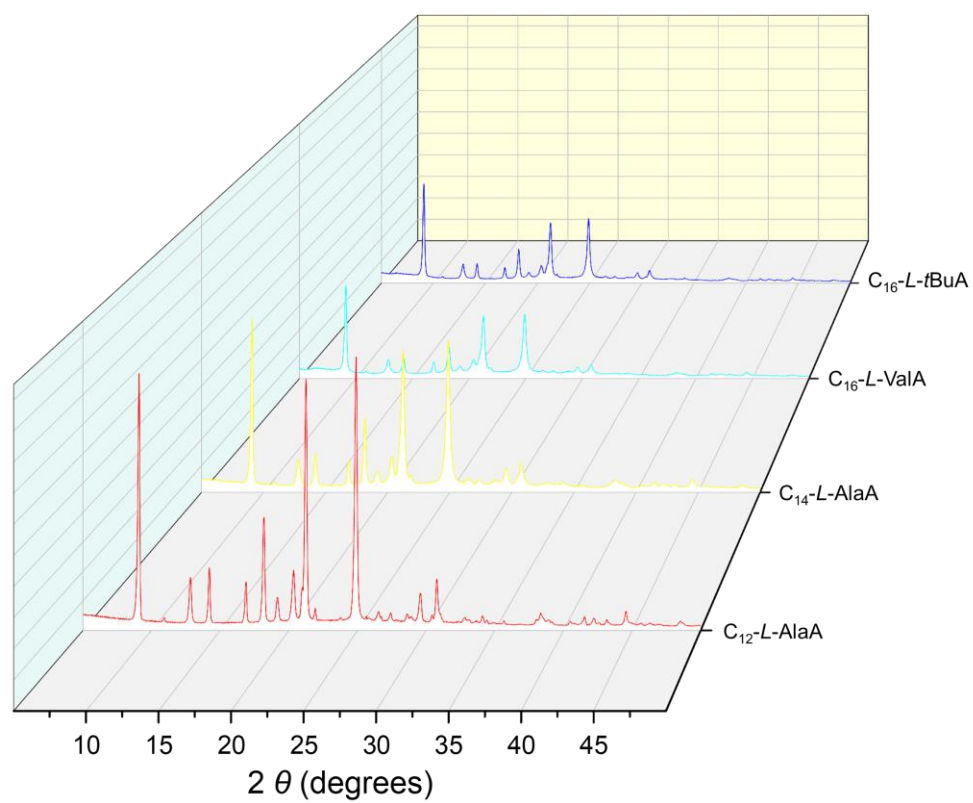


Figure S15. Powder XRD patterns of conformational chiral COF-300 synthesized using different amino-acid derivatives

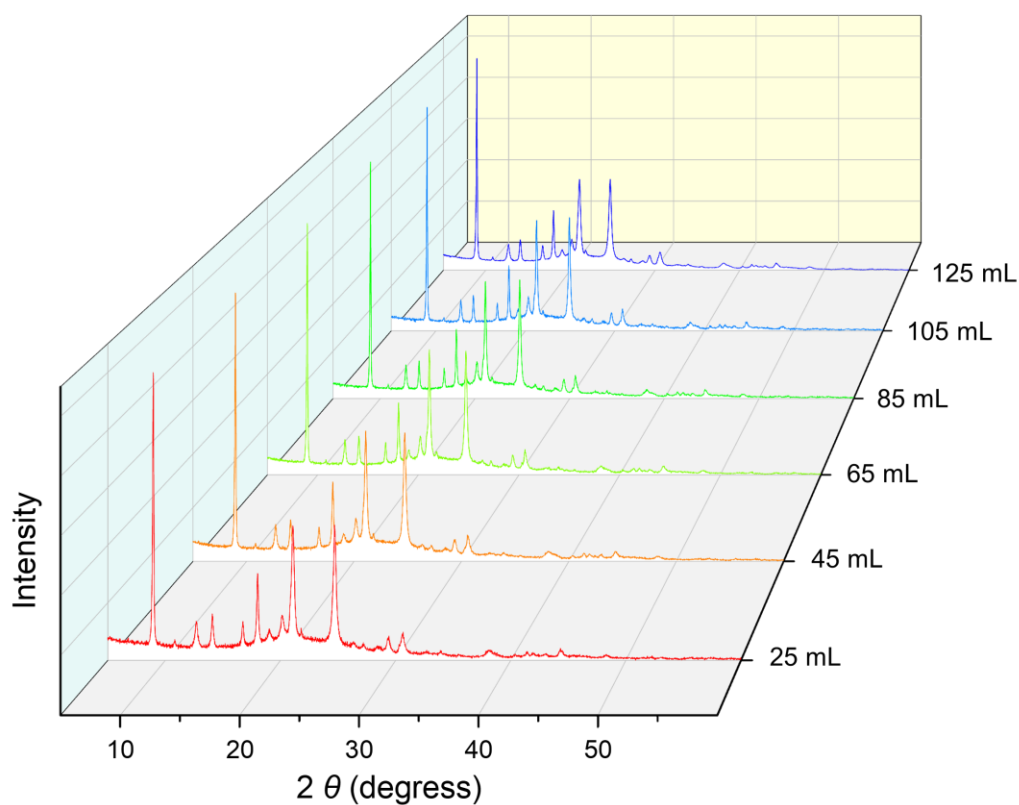


Figure S16. Powder XRD patterns of conformational chiral COF-300 under varying water concentrations.

Table S12. Unit cell parameters of COF-300_L-cp were analyzed under varying water concentrations (25 mL, 45 mL, 65 mL, 85 mL, 105 mL, and 125 mL).

Water	$(a = b)/\text{\AA}$	$c/\text{\AA}$	Volume/ $\text{\AA}^3$	$R_{wp}/\%$	$R_p/\%$
25 mL	19.720(3)	8.872(1)	3450(1)	4.25	3.30
45 mL	19.702(1)	8.888(1)	3450.0(5)	5.53	4.15
65 mL	19.670(1)	8.907(1)	3446.3(7)	5.49	4.04
85 mL	19.643(1)	8.9159(9)	3440.1(3)	7.67	5.87
105 mL	19.625(1)	8.9221(6)	3436.4(2)	6.26	4.65
125 mL	19.600(1)	8.9332(6)	3431.9(4)	6.03	4.29

Section 4. Optical microscope images

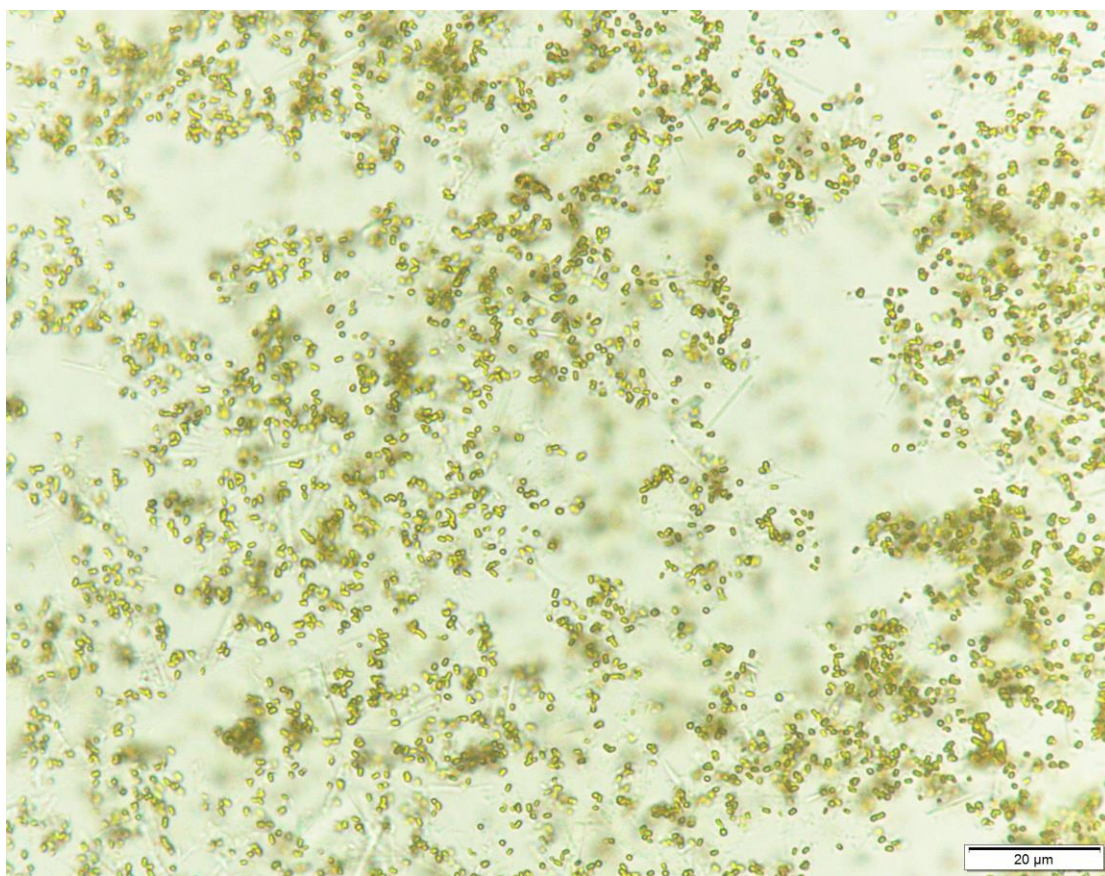


Figure S17. Optical microscope image of chiral COF-300 synthesized from the C₁₆-L-AlaA template.

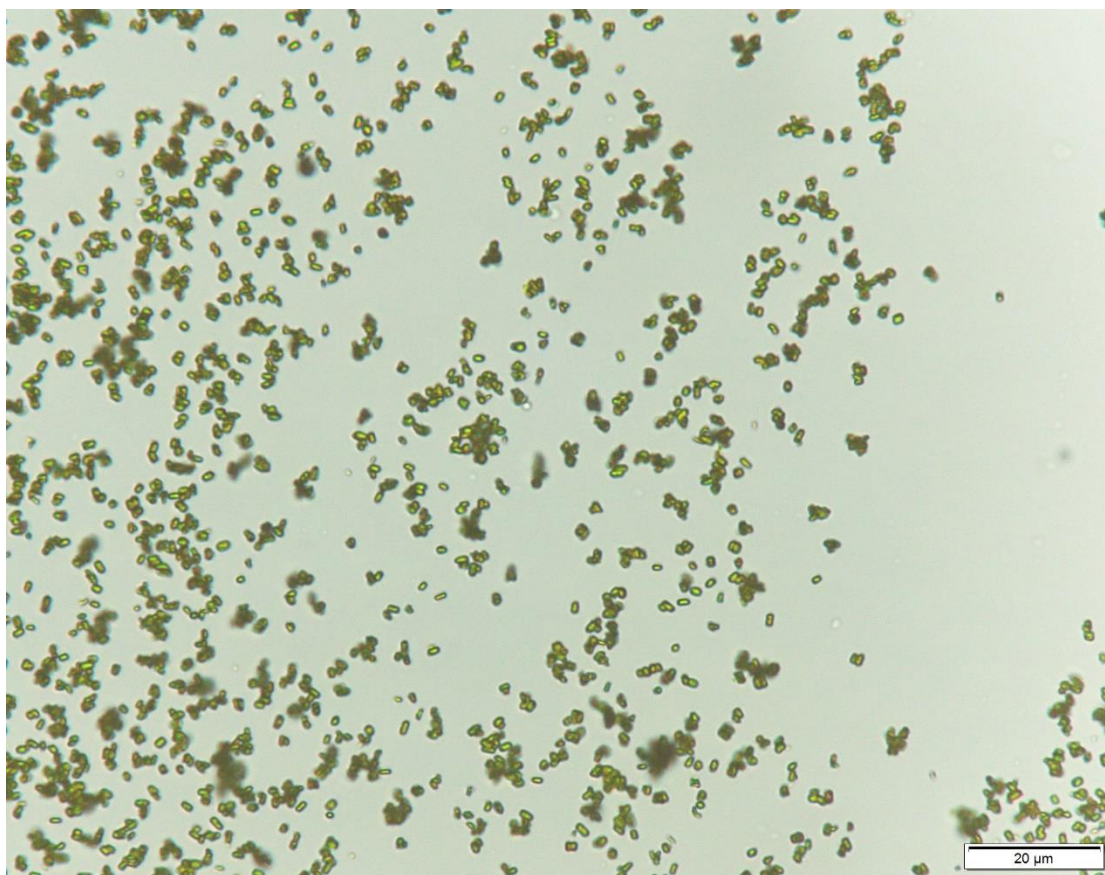


Figure S18. Optical microscope image of chiral COF-300 synthesized from the C₁₆-D-AlaA template.

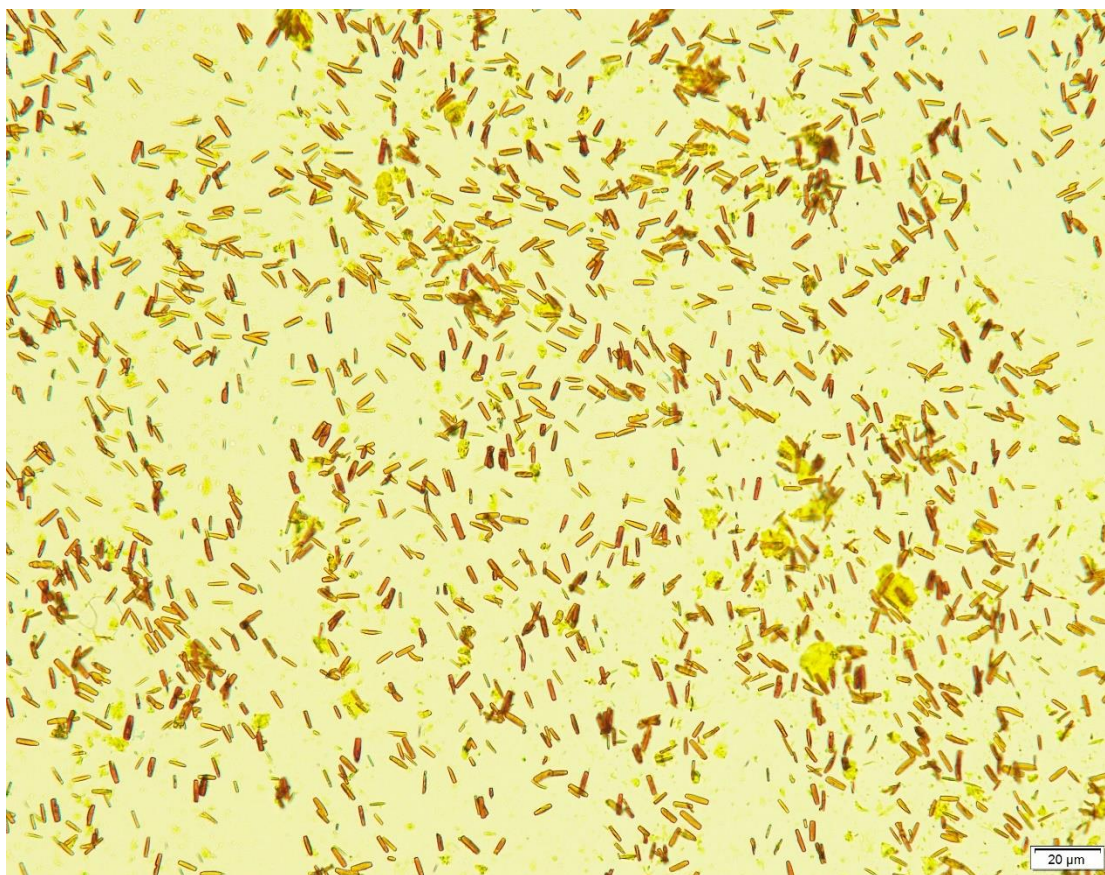


Figure S19. Optical microscope image of chiral COF-304.

Section 5. Thermal stability analysis

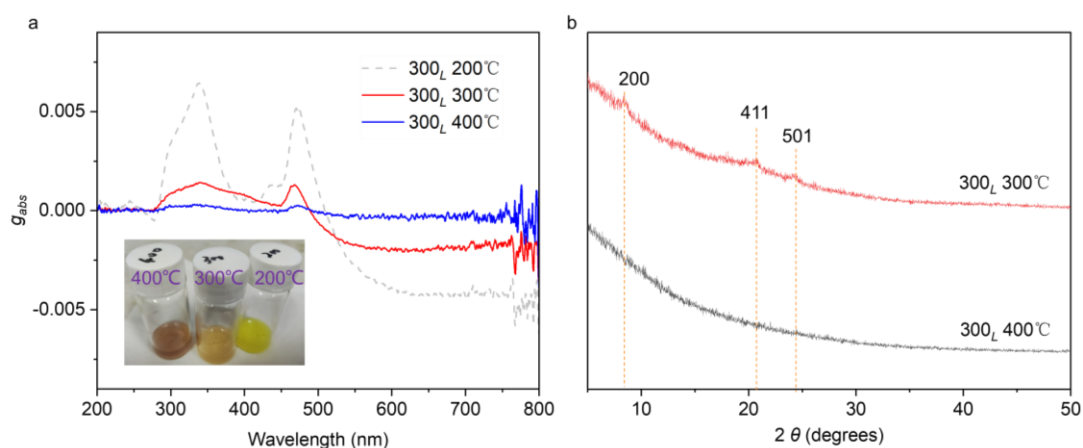


Figure S20 **a, b**, Circular dichroic (CD) absorption spectra and powder X-ray pattern of COF-300_L dispersed in water after treatment at 300 °C and 400 °C, respectively.

We investigated higher annealing temperatures by dispersing COF-300_L in water after annealing at 300 °C and 400 °C, followed by circular dichroic (CD) absorption spectral analysis (Figure S20a). The results reveal a significant decrease in signal intensity at 300 °C compared to 200 °C, with almost no chiral signal observed at 400 °C. Corresponding PXRD analysis shows that diffraction intensity is already quite weak at 300 °C and nearly absent at 400 °C (Figure S20b), indicating that higher temperatures induce racemization and structural degradation.

Section 6. Crystal size distribution analysis

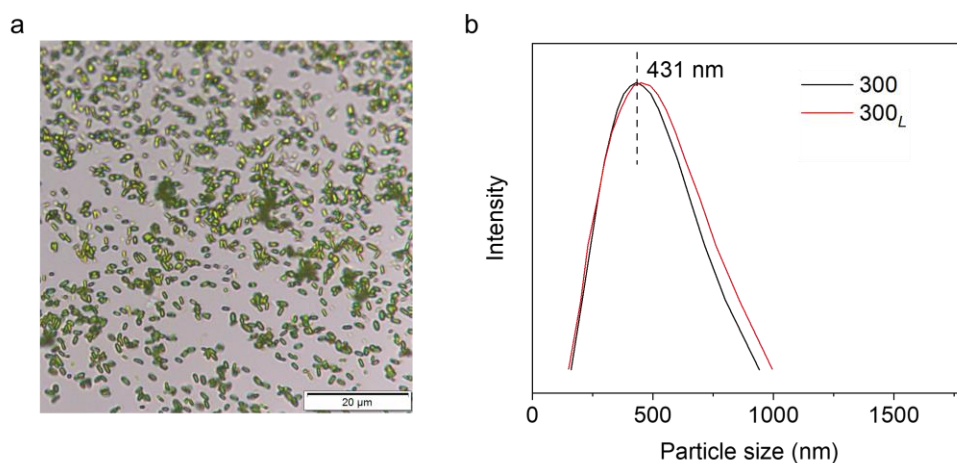


Figure S21 **a**, Optical microscope image of achiral COF-300. **b**, Particle size distribution of COF-300 and COF-300_L dispersed in water.

By comparing optical microscope images of achiral COF-300 (Figure S21a) and chiral COF-300_L (Supplementary Figure S17), we observed that both materials consist of monodisperse single-crystal particles. Dynamic light scattering measurements of the samples dispersed in water show an average particle size of approximately 431 nm for both (Figure S21b), suggesting that the observed differences in CO₂ adsorption are not attributable to size effects³².

Section 7. Infrared spectral analysis

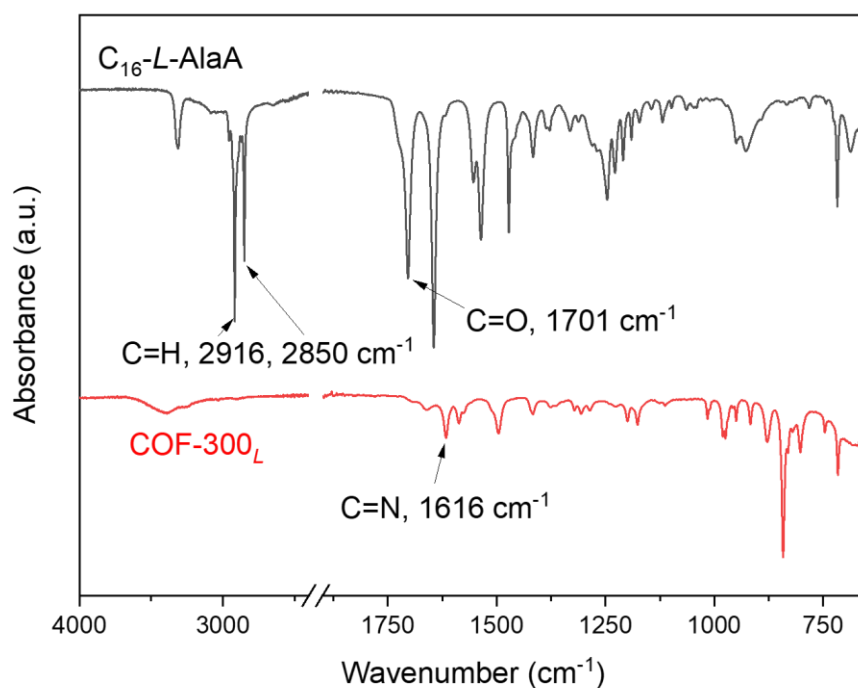


Figure S22 Infrared spectra of C_{16} -L-AlaA and COF-300_L-cp.

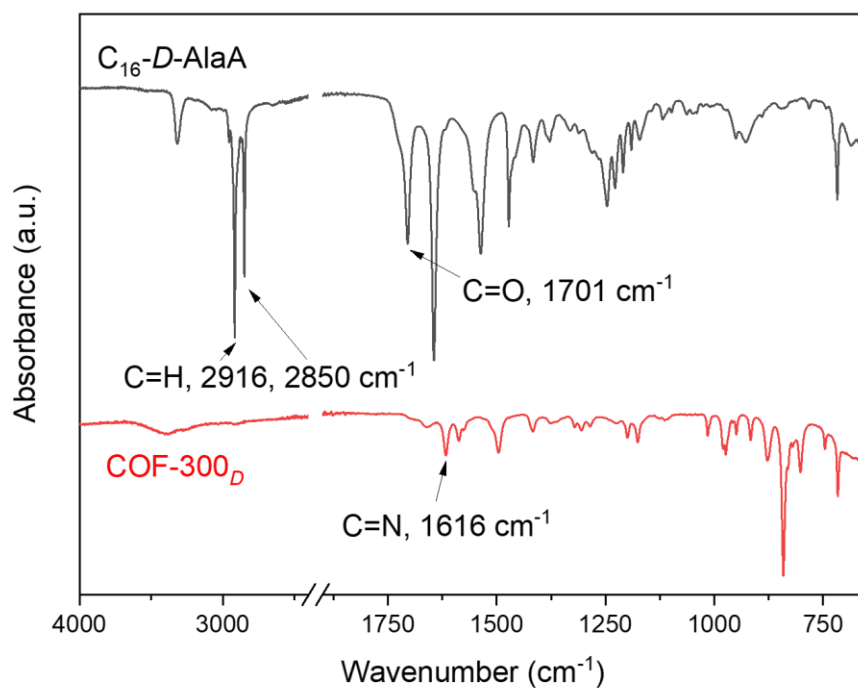


Figure S23 Infrared spectra of C_{16} -D-AlaA and COF-300_D-cp.

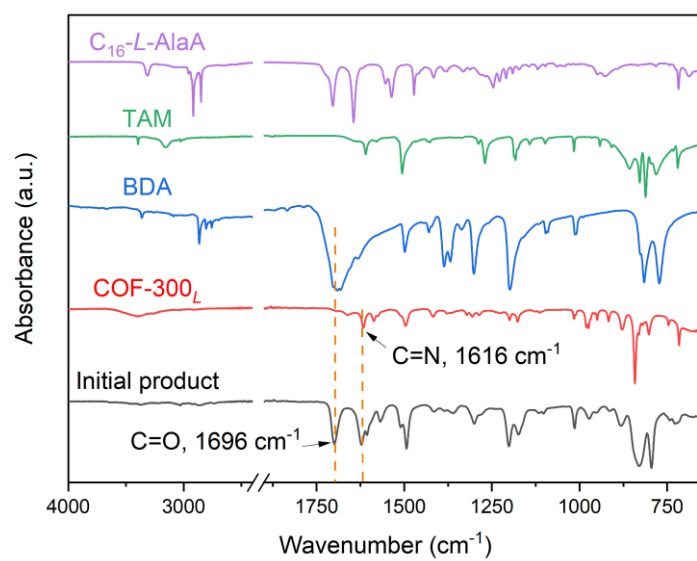


Figure S24. Infrared spectra of initial product, COF-300_L, BDA, TAM and C₁₆-L-AlaA.

Section 8. Solid-state nuclear magnetic resonance spectroscopic analysis

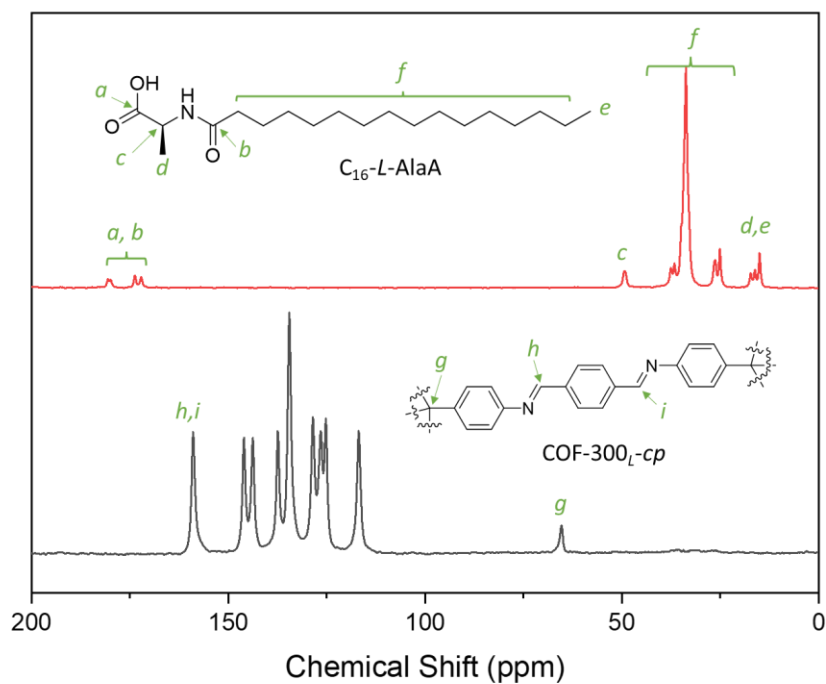


Figure S25 ^{13}C CP/MAS spectrums of $\text{C}_{16}\text{-L-AlaA}$ and $\text{COF-300}_\text{L-cp}$.

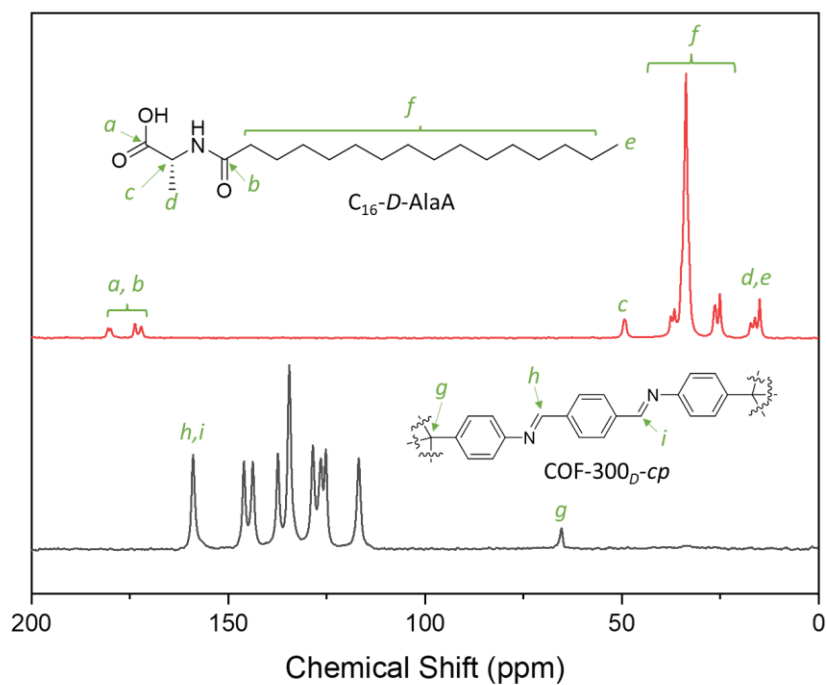


Figure S26 ^{13}C CP/MAS spectrums of $\text{C}_{16}\text{-D-AlaA}$ and $\text{COF-300}_\text{D-cp}$.

Supplementary References

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